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Habib University

EE-437 Renewable energy systems – Fall 2025

Time = 15 minutes

Quiz 06 – CLO 4

Max Points: 20

CLO 4 - 20 marks

Question 1 [3 points]: Differentiate/ define DSO, TSO and IPP.

DSO: Distribution of electric supply in cities

TSO: Transmission of electric supply inter city and even between countries

IPP: Independent power producer of electric supply

Question 2 [3 points]: Define energy markets, capacity markets and ancillary services markets.

Energy: Contracts based on actual energy supplied to the grid

Capacity: Contracts based on keeping available power without supplying energy

Ancillary: Grid auxiliary services such as reactive power control and frequency control

Question 3 [2 points]: Explain merit order in 2 sentences.

All power stations are placed in order of their cost of producing energy and capacity. The highest cost power station inside the demand sets electricity price.

Question 4 [2 points]: What are the two methods of demand side integration?

DSM: remotely manage load

DSR: active response of consumers

Question 5 [2 points]: How can changing a solar panel's orientation help the grid?

By moving to southwest orientation more energy is produced during sunset when energy is in higher demand. Different orientations give smoother daily production.

Question 6 [2 points]: How does a low wind turbine help the grid?

Low speed turbines give more output when winds are slow compared to a classical turbine and less power when winds are high. This gives a smoother output of power.

Question 7 [4 points]: Define microgrid and supergrid and how are they important for renewable energy integration?

Microgrid: A microgrid is a small, localized power system with its own generation and storage that can operate with or without the main grid, helping integrate renewable energy by balancing local variability and improving resilience.

Supergrid: A supergrid is a large, long-distance transmission network that links multiple regions to transfer renewable energy over wide areas, enabling large-scale balancing and reducing variability in supply.

Question 8 [1 point]: What does reactive power have to do with voltage in the grid?

Reactive power is directly tied to voltage control in the grid: supplying reactive power raises voltage, while absorbing reactive power lowers voltage. This is due to dominance of reactive components in HVAC setups.

Question 8 [1 points]: A generator provides 0.5MVA reactive power and 0.5MW. What is the apparent power it provides?

$$S = \sqrt{P^2 + Q^2} = \sqrt{0.5^2 + 0.5^2} = 0.707MVA$$